Gowin DDR3 Memory Interface IP

User Guide

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Revision History

Date	Version	Description
08/24/2018	1.0E	Initial version published.
03/12/2019	1.1E	Description of DDR3 1:4 clock ratio about continuous burst added.
07/12/2019	1.2E	● Address format modified. ● Read and write efficiency description added.
01/06/2020	1.3E	Static parameter options modified.
06/30/2021	1.4E	● memory_clk and pll_lock ports added. ● Design source code file updated.
11/25/2021	1.5E	● The timing descriptions of cmd_en and cmd_ready modified. ● The timing descriptions of wr_data_en and wr_data_rdy modified.
08/12/2022	1.6E	● The timing descriptions of user interface updated. ● The descriptions for continuous burst mode and non-continuous burst mode of the controllers added.
10/17/2022	1.7E	wr_data_wren modified to wr_data_en, clk modified to clk_out, and wr_data_ready modified to wr_data_rdy in the timing diagrams.
06/08/2023	2.0E	● The description of GW5AST-138 device added. ● pll_stop signal added. ● Chapter 8 Reference Design and Chapter 9 File Delivery removed.
08/18/2023	2.1E	● GW5A-25 devices added. ● Burst_Number_Enable option removed. ● The descriptions of app_burst_number interface signals removed.
09/12/2023	2.2E	GW5A-25 devices support pll_stop.
12/31/2024	2.3E	● The maximum rate supported by devices and related descriptions modified. ● The content related to pll_stop of the IP in GW5A(T)-60K environment added. ● Data_Mask port description updated. ● Descriptions of reset updated.
03/14/2025	2.4E	● Descriptions of IP features updated. ● Descriptions of resource utilization updated.
05/09/2025	2.5E	● Descriptions of ODT added. ● Descriptions of static parameters updated.
01/16/2026	2.6E	● Description of the AXI4 interface added. ● Description of the "Buffer Type" option added.

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1 About This Guide

1.1 Purpose

The purpose of Gowin DDR3 Memory Interface IP User Guide is to help you quickly learn the features and usage of Gowin DDR3 Memory Interface IP by providing the descriptions of the functions, ports, timing, GUI, and reference design. The software screenshots in this manual are based on V 1.9.12.01. As the software is subject to change without notice, some information may not remain relevant and may need to be adjusted according to the software that is in use.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI Website. You can find the related documents at www.gowinsemi.com:

- DS102, GW2A series of FPGA Products Data Sheet
- DS226, GW2AR series of FPGA Products Data Sheet
- DS1104, GW5AST series of FPGA Products Data Sheet
- DS1103, GW5A series of FPGA Products Data Sheet
- SUG100, Gowin Software User Guide

1.3 Terminology and Abbreviations

The terminology and abbreviations used in this manual are as shown in Table 1-1.

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Meaning
ECC	Error Correcting Code
FIFO	First Input First Output
GSR	Global System Reset
IP	Intellectual Property
LUT	Look-up Table
ODT	On-Die Termination
RAM	Random Access Memory

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly by the following ways.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 Overview

Gowin DDR3 Memory Interface IP is a commonly used DDR3 memory interface IP that complies with the JESD79-3F standard protocol. The IP design includes DDR3 memory controller (MC) and the corresponding physical interface (PHY). Gowin DDR3 Memory Interface IP provides you with a general command interface which can connect with the memory chip to access and save data.

Table 2-1 Gowin DDR3 Memory Interface IP

Gowin DDR3 Memory Interface IP	
Logic Resource	Please refer to Table 3-1.
Delivered Doc.	
Design Files	Verilog (encrypted)
Reference Design	Verilog
TestBench	Verilog
Test and Design Flow	
Synthesis Software	GowinSynthesis
Application Software	Gowin Software (V1.9.11 and above)

Note!

For the devices supported, you can click here to get the information.

3 Features and Performance

3.1 Features

- Interfaces to the industrial standard DDR3 SDRAM devices and modules that are compatible with the JESD79-3F specification
- Supports memory data path width of 8 bits, 16 bits, 24 bits, 32 bits, 40 bits, 48 bits, 56 bits, 64 bits, and 72 bits
- Supports the single row RDIMM, UDIMM, and SODIMM memory modules
- Supports x8 and x16 data width memory chips
- Programs 4, 8, or OTF burst lengths
- Supports 1:2 and 1:4 clock ratio for GW2A-18, GW2A-55, and GW2AR-18, and supports 1:4 clock ratio for GW5AST-138
- Supports ECC
- Configurable CL
- Configurable AL
- Configurable CWL
- Configurable t_{FAW}
- Configurable t_{RAS}
- Configurable t_{RCD}
- Configurable t_{RFC}
- Configurable t_{RRD}
- Configurable t_{RTP}
- Configurable t_{WTR}
- Supports dynamic on-chip ODT
- Supports automatic refreshing and user startup refreshing and the interval of automatic refreshing is configurable
- Supports AXI4 interface

3.2 Operating Frequency and Bandwidth Efficiency

The data rate of the DDR3 SDRAM that Gowin DDR3 Memory Interface supports is as follows:

- In 1:2 clock ratio mode, the maximum rate is 533 Mbps.
- In 1:4 clock ratio mode:
  - For GW5A(R)(S)-25 devices, the maximum rate is 1100Mbps.
  - For GW5A(T)-60 devices, the maximum rate is 1100Mbps.
  - For other devices, the maximum rate is 800Mbps.

The bandwidth efficiency of Gowin DDR3 Memory Interface IP is as follows:

- In 1:2 clock ratio mode with a burst length of 4, the bandwidth efficiency is 50%.
- In 1:2 clock ratio mode with a burst length of 8, the bandwidth efficiency is 90%.
- In 1:4 clock ratio mode, the bandwidth efficiency is 90%.

Gowin DDR3 Memory Interface IP employs the Verilog language, and it is applied to GW2A and GW5A (excluding 15K) series chips. Due to differences in device density and speed, resource utilization may vary. Taking the GW5A-LV25UG324C2/I1 chip as an example, the resource utilization is as shown in Table 3-1.

Table 3-1 Resource Utilization (1)

Device	Speed Grade	Name	Resource Utilization
GW5A-25	C2	LUT	2173
		REG	3410
		ALU	125
		BSRAM	8

In Table 3-1, Gowin DDR3 Memory Interface is configured as a custom user interface with a user address width of 28 bits, DRAM width of x16 and the clock ratio of 1:4.

If it is configured as an AXI4 interface, the resources utilized by the AXI4 interface are shown in Table 3-2.

Table 3-2 Resource Utilization (2)

Device Series	Speed Grade	Name	Resource Utilization
GW5A-25	C2	LUT	252
		REG	168
		ALU	35
		BSRAM	4

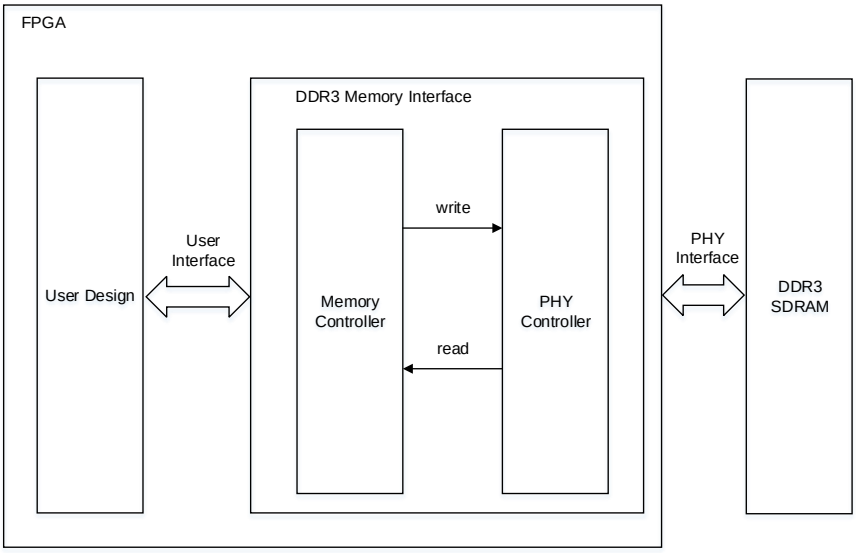
In Table 3-2, the AXI4 interface is configured with a 28-bit address width, a 128-bit data width, and a 4-bit ID width, with internal buffering using BSRAM resources. If the buffer is configured as SSRAM, the logic resource will be increased. Users need to configure this setting themselves and synthesize the design using Gowin Software to check the actual resource utilization.

4 Functional Description

4.1 Structure

The structure of Gowin DDR3 Memory Interface IP is as shown in Figure 4-1. It includes Memory Controller and Physical Interface modules, etc. The User Design in the Figure 4-1 is the user design in the FPGA that needs to be connected to an external DDR3 SDRAM chip.

Figure 4-1 DDR3 Memory Interface IP Structure Diagram



4.2 Memory Controller

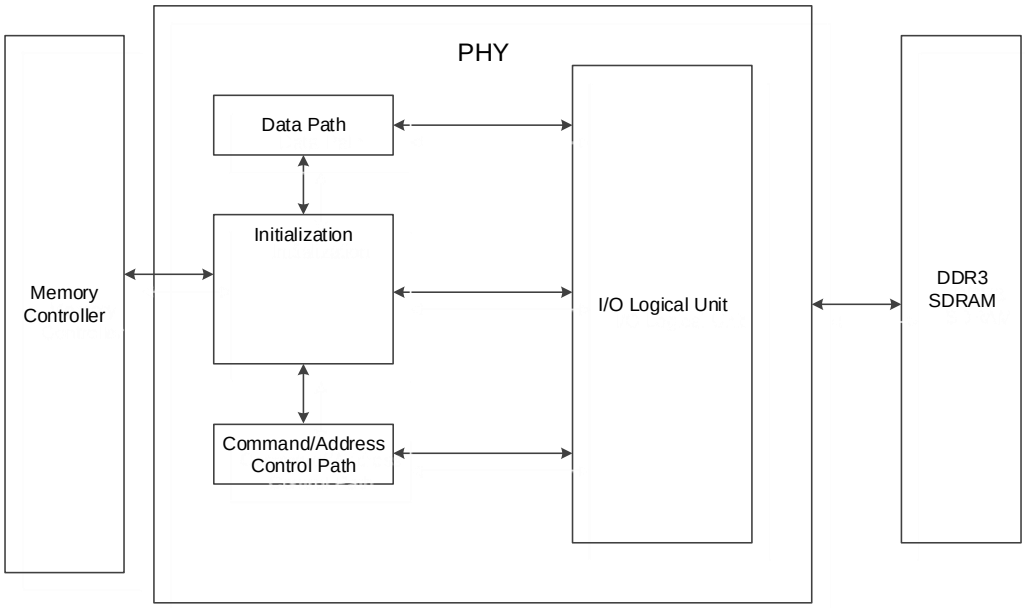
Memory Controller is located at MC layer, realizing protocol layer functions. Internal state machine is for BANK, ROW, COL, and refresh control. Memory Controller receives user-side read and write commands, stores the commands in FIFO logic internally, converts read and write commands into detectable interface timings on PHY side, and inputs to PHY side.

4.3 PHY

The PHY provides the physical definition and interface between MC and external DDR3 SDRAM, receives commands from the memory controller on MC layer, and provides the DDR3 SDRAM chip with interface timing.

The structure of PHY includes four modules: initialization module, data path, command address control path, and I/O logic module, as shown in Figure 4-2.

Figure 4-2 DDR3 PHY Basic Structure Diagram



4.3.1 Initialization Unit

The initialization module is mainly used for initialization and read calibration after DDR3 SDRAM power-on. After initialization and read calibration are finished, the "init_calib_complete" signal will be high from low, indicating that the initialization has been completed.

Note!

Read/write operations are not allowed to be performed until the init_calib_complete signal is pulled up.

Power-on Initialization

According to the JESD79-3F protocol, there is a need to initialize the DDR3 SDRAM (chip or DIMM) after power-on. This includes the reset, clock enable, mode register configuration, and ZQ calibration.

4.3.2 Data Path Unit

Data path includes write data path and read data path.

4.3.3 Control Path Unit

The command/address control path is a single path that receives the command and address signal sent by the Memory Controller and cooperates with the data path to process the write and read data delay

parameters and sends the commands to the I/O logic module.

4.3.4 I/O Logical Unit

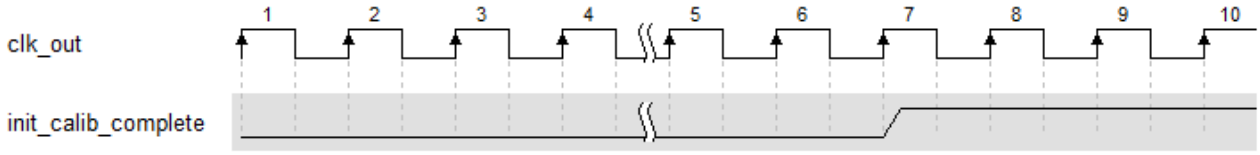
The Logic I/O module is mainly used to convert the clock domain of the data, command, and address signals received from the data path and command/address path.

4.4 User Interface

4.4.1 Initialization Interface

Init_calib_complete: The DDR3 SDRAM needs to be calibrated before write and read. Therefore, after power-on, PHY will initialize and calibrate the DDR3 SDRAM, and the init_calib_complete is pulled high after the initialization is completed, as shown in Figure 4-3.

Figure 4-3 Initialization Completion Signal Timing Diagram

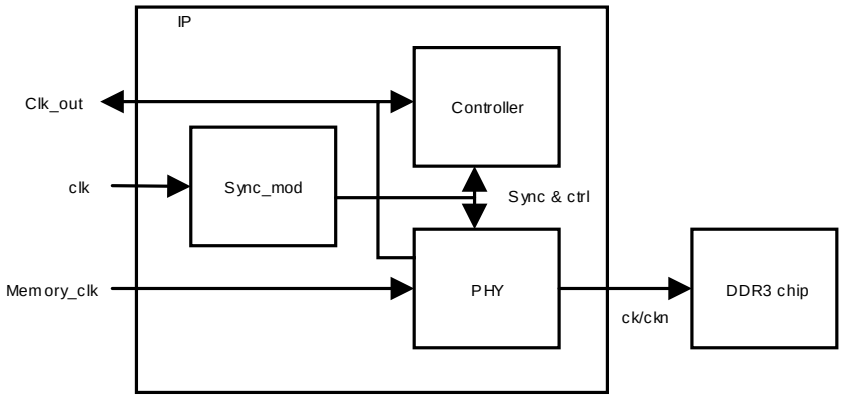


4.4.2 Clock

The IP has three clocks, two input clocks, clk and memory_clk, and one output clock, clk_out.

As shown in Figure 4-4, clk is used to generate some synchronization and control signals which act on the main logic of the IP (PHY layer and Memory controller). clk is required to be a low-speed continuous clock with a recommended value of 50MHz, and the input of the on-board crystal can be connected to clk.

Figure 4-4 Clock



memory_clk is the high speed clock, which uses HCLK resources to drive the PHY and is output to the DDR3 chip.

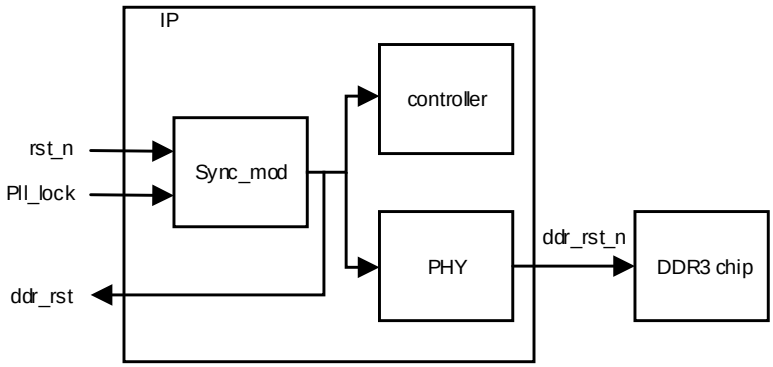
clk_out is the frequency-divided clock of memory_clk. When clk_ratio is set to 4:1, clk_out is the four-divided clock of memory_clk; when clk_ratio=2:1, clk_out is the two-divided clock of memory_clk, and it uses PCLK resources. clk_out is used as the logic processing clock for IP and

is output to user logic. The user's interface operation on the IP should be synchronized with clk_out.

4.4.3 Reset

The IP includes the input signals rst_n and pll_lock, as well as the output signal ddr_rst. As shown in Figure 4-5, rst_n is required to do a synchronization release under the clk before sending it to the IP. rst_n will reset the Sync_mod module logic, and then Sync_mod waits for pll_lock. After pll_lock is locked, Sync_mod will output ddr_rst as the global reset of the IP and send it to the user. Any reset logic can be connected to rst_n, and pll_lock can only be connected to the LOCK signal of the PLL. If pll_lock is not connected to the LOCK signal of the PLL, the IP will not be able to detect whether the clock is stable, and DDR initialization failure is prone to occur.

Figure 4-5 Reset

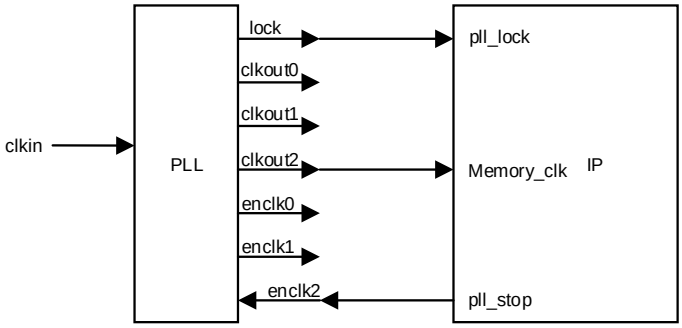


4.4.4 pll_stop

pll_stop is a control signal for GW5A(S)(T) devices; it is a switch to control memory_clk, active-low.

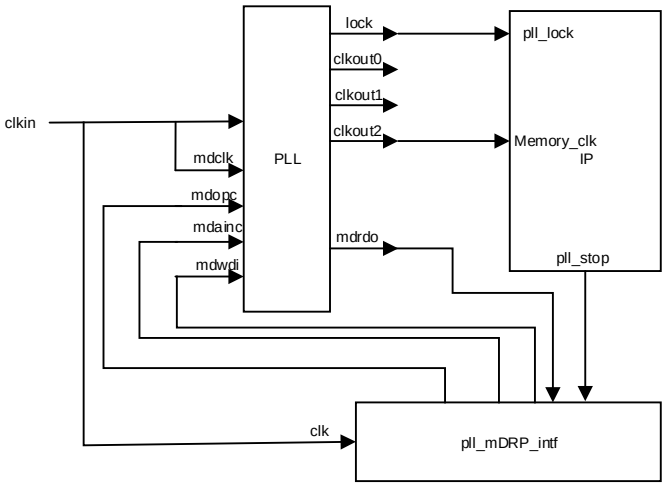
As shown Figure 4-6, when 138K devices are used, pll_stop is directly connected to enclk2 of PLL.

Figure 4-6 138K pll_stop



As shown in Figure 4-7, when 25K devices are used, pll_stop needs to be connected to the adapter module, pll_mDRP_intf, to indirectly control the clkout2 of PLL. The clk of pll_mDRP_intf is the same as the mdclk and clkin of PLL.

Figure 4-7 25K pll_stop



For other GW5A series of devices, the usage of pll_stop will not be explained in details. It is necessary to check whether the target FPGA's PLL has an enable control enclk similar to that of the 138K. If it does, pll_stop can be directly connected. Otherwise, control must be implemented via the mDRP interface.

pll_mDRP_intf module is included in the reference design of Gowin_DDR3_Memory_Interface_RefDesign, please download it from official website.

4.4.5 Command and Address Interface

Command

You can write commands to IP via cmd and cmd_en, and the controller initiates read and write to the DDR3 chip according to the order of the write commands.

- Cmd is the command port;
- Cmd_en is the enable signal of the command, active-high.

See Table 4-1 for the meanings of cmd：

Table 4-1 Cmd

Command	cmd[2:0]
Read	3’b001
Write	3’b000

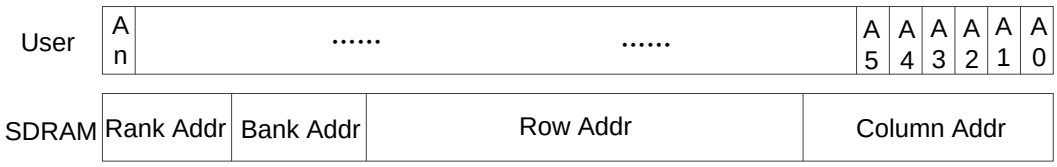
Address

Addr is the user-side address bus which is written to the controller together with cmd. Addr is valid when cmd_en is valid.

In the application, a mapping relationship exists between the address bus of user interfaces and Bank, Rank, Row, Column of the physical memory. In this design, the array is in Rank-Bank-Row-Column order, and the addressing scheme is as shown in Figure 4-8. You should note the

order of the addresses provided in the application.

Figure 4-8 Addressing Scheme in Rank-Bank-Row-Column Order



Addr is the DDR address, i.e. addr directly reflects the DDR memory address. When DDR3 burst_mode is configured with BC4, one write/read needs to write/read 4 dq data to DDR, so one DDR write/read occupies 4 addresses. When DDR3 burst_mode is configured with BL8, one write/read needs to write/read 8 dq data to DDR, so one DDR write/read occupies 8 addresses. You should note the control of the addresses provided in the application.

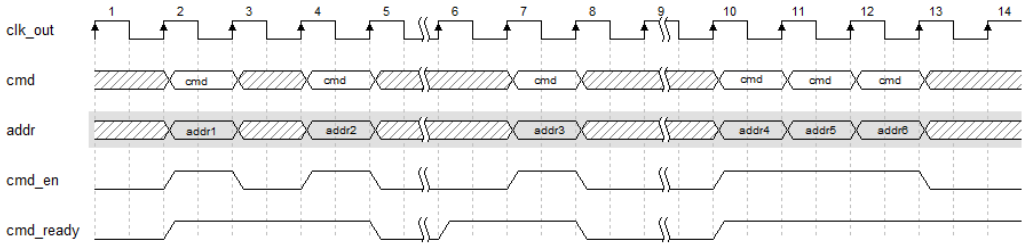
Address and Command Timing with 1:2 Clock Ratio

If clock ratio is 1:2 and cmd_ready is high, it indicates that the DDR controller can receive user commands.

When cmd_en and cmd_ready is 1, write cmd and addr to IP. addr1 and addr2 have no relationship and do not have to be neighboring addresses.

The timing of the command, address, and enable signals is as shown in Figure 4-9.

Figure 4-9 Command, Address, and Enable Signal Timing-A



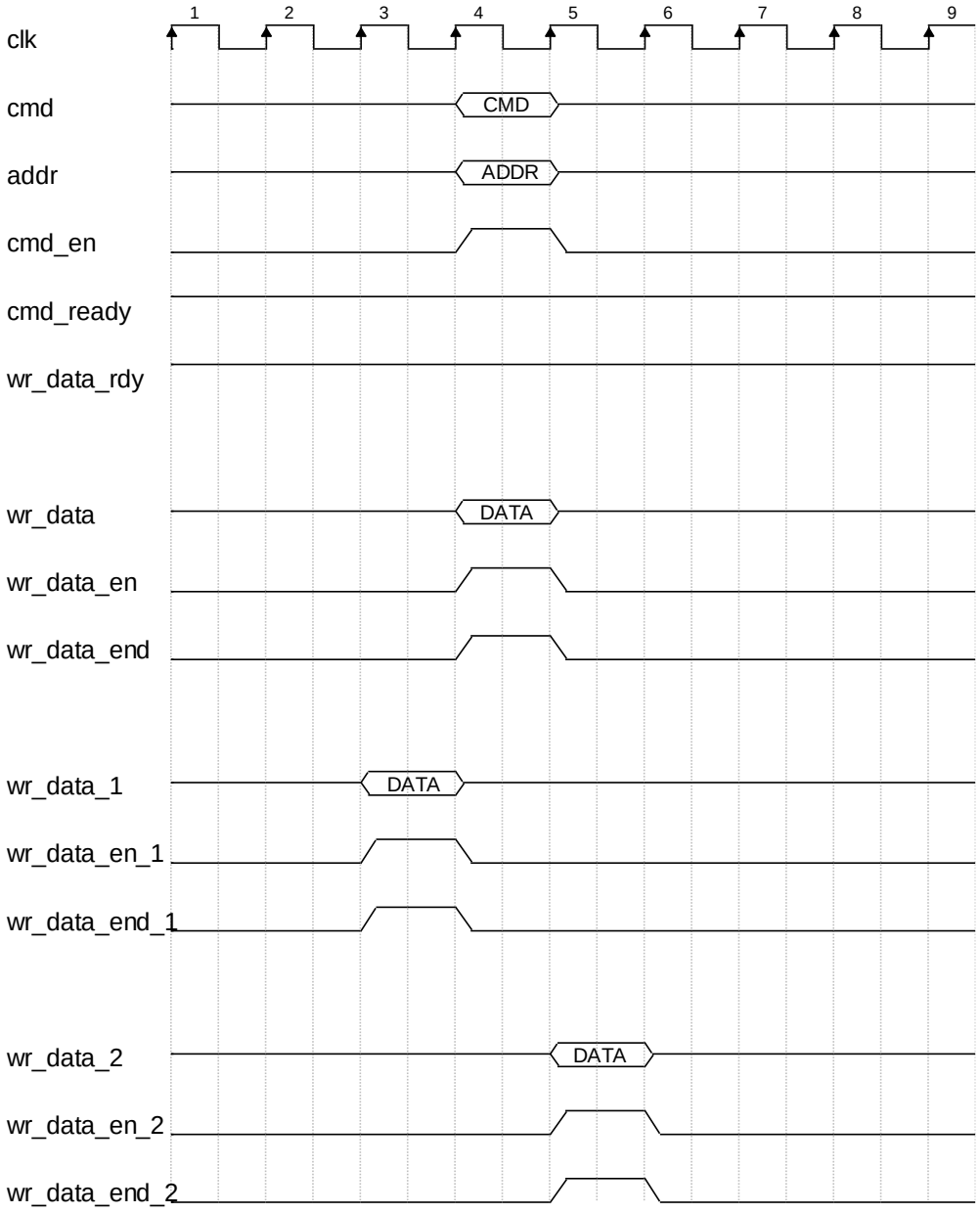
Address and Command Timing with 1:4 Clock Ratio

The address and command timing is the same as that in 1:2 clock ratio. See the timing as shown in Figure 4-9.

4.4.6 The Location Relationship Between Command and Write Data

The location relationship between command and write data is as shown in Figure 4-10.

Figure 4-10 The Location Relationship Between cmd and Data



4.4.7 Write Data Interface

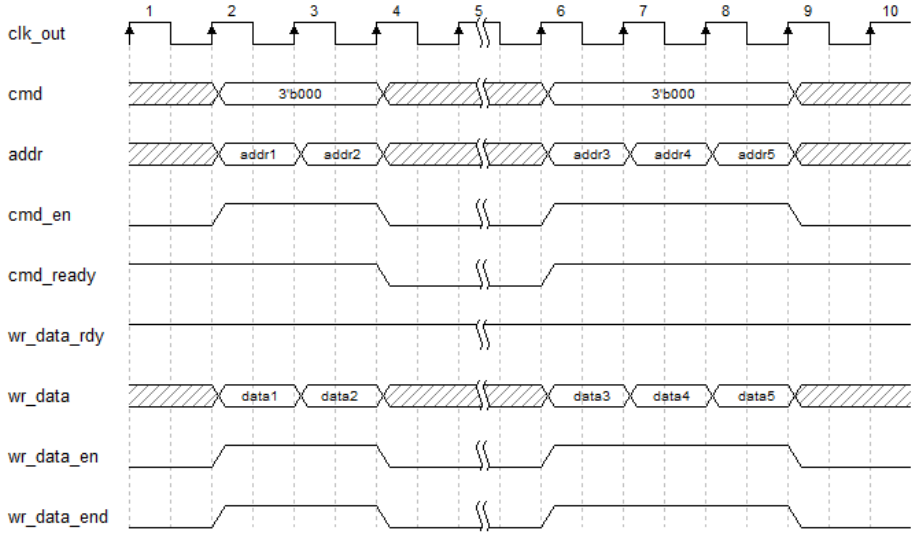
- wr_data: Data bus interface. You can write the data need to be stored in DDR through this interface.
- wr_data_en: Data write enable interface. wr_data is active-high.
- wr_data_end: It indicates the data on wr_data bus in current cycle is the last write data currently.
- wr_data_rdy: when wr_data_rdy is high, it indicates the controller can receive user data. You can write data in controller through wr_data, wr_data_en, and wr_data_end interfaces.

When the clock ratio is 1:2, the burst_mode is configured with BC4, and the bit width ratio of wr_data to dq data is 1:4. At this time, one wr_data can meet one burst write for DDR. wr_data_en and wr_data_end

have the same action, so you can write 1 to wr_data_en and wr_data_end at the same time when writing data.

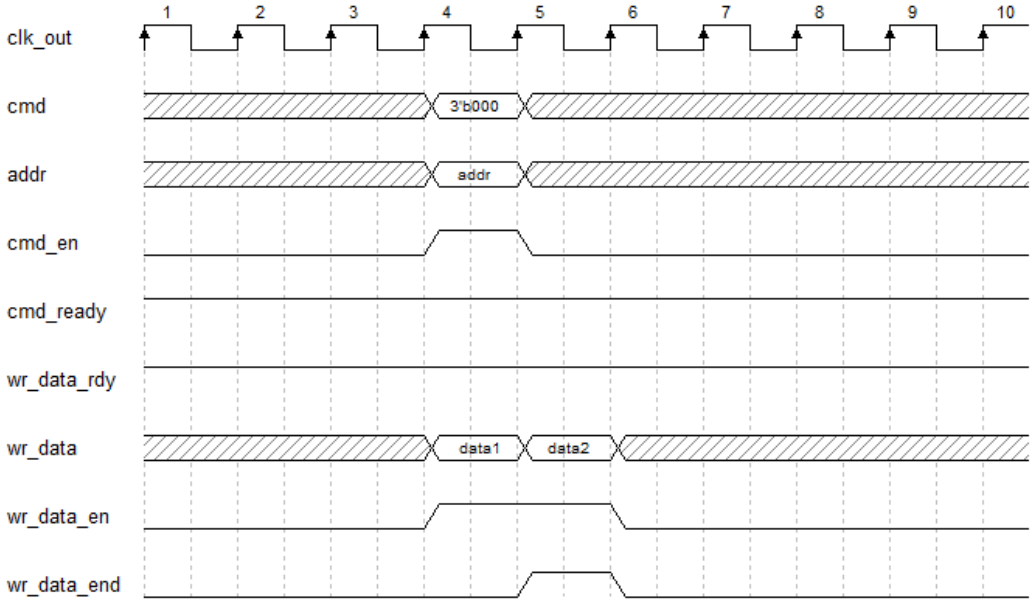
When the clock ratio is 1:4 and the burst_mode is configured with BL8, the situation is the same as the above example. See Figure 4-11 for the timing.

Figure 4-11 Write Data Timing with 1:2 Clock Ratio and Burst_Mode=BC4 or 1:4 Clock Ratio



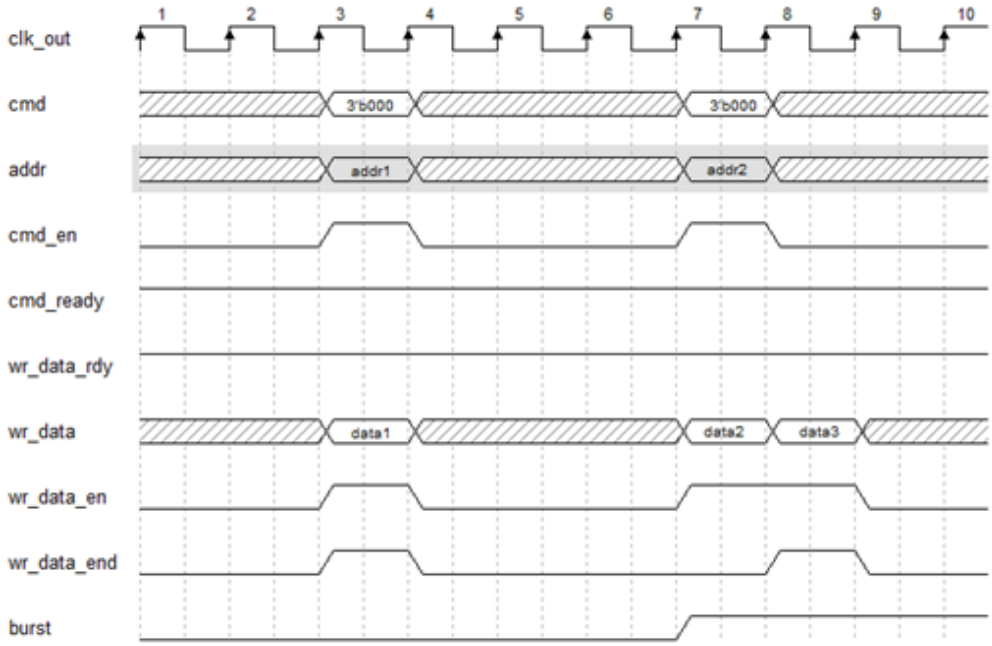
When the clock ratio is 1:2 and burst_mode is configured with BL8, the bit-width ratio of wr_data to dq data is 1:4. At this time, DDR needs two wr_data for one burst write. Then wr_data_en should last for two consecutive cycles, and the second cycle will write wr_data_end to 1. See Figure 4-12 for the timing.

Figure 4-12 Write Data Timing with 1:2 Clock Ratio and Burst Mode=BL8



Only when the clock ratio is 1:2, burst_mode is supported to be configured with OTF mode. In this mode, the port signal burst is 0 indicating the current DDR chip burst_mode is BC4; the port signal burst is 1 indicates the current DDR chip burst_mode is BL8. At this time, you should control the wr_data _en and wr_data_end signals according to the situation. See Figure 4-13 for the timing.

Figure 4-13 Write Data Timing with 1:2 Clock Ratio and Burst Mode=OTF



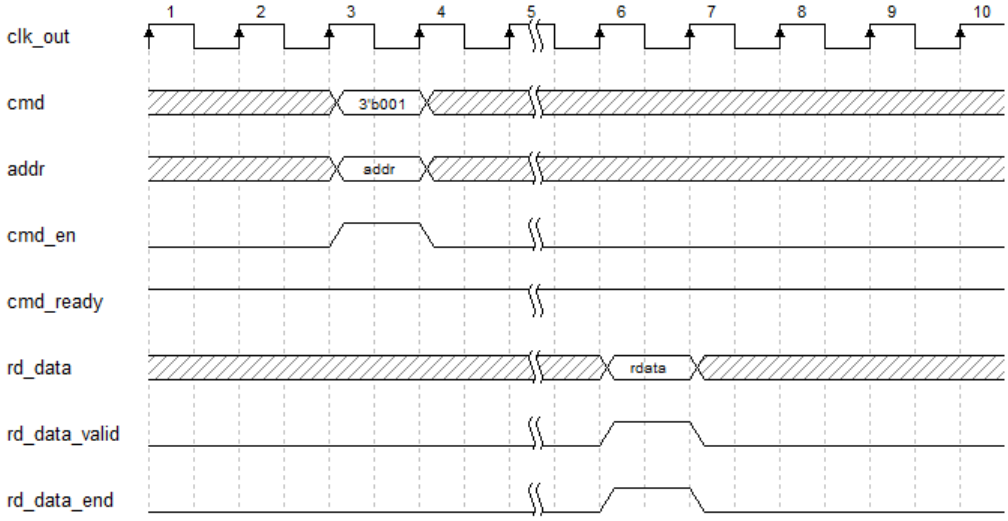
4.4.8 Read Data

You can read the data from the DDR3 SDRAM using the user interfaces, including rd_data, rd_data_valid, and rd_data_end.

- The rd_data is the returned read data port.
- The rd_data_valid port is the valid read data port. When it is high, it indicates the returned rd_data is valid at this time.
- The rd_data_end port indicates the last set of returned data in the current burst_mode, active-high.

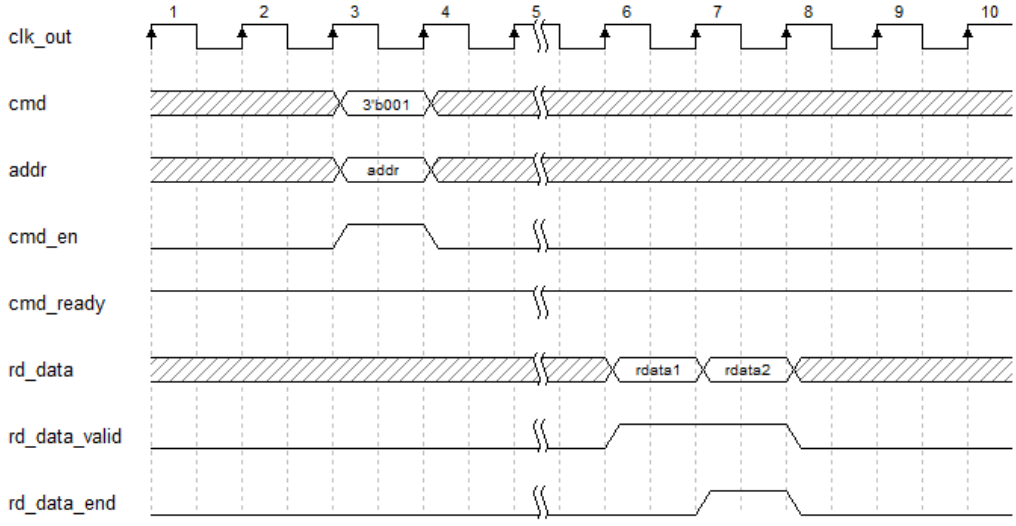
Similar to the write operation, when clock ratio is 1:2 and users configure the burst length with BC4, read data occupies one clk cycle, as shown in Figure 4-14.

Figure 4-14 Read Data Timing with 1:2 Clock Ratio and Burst Mode=BC4



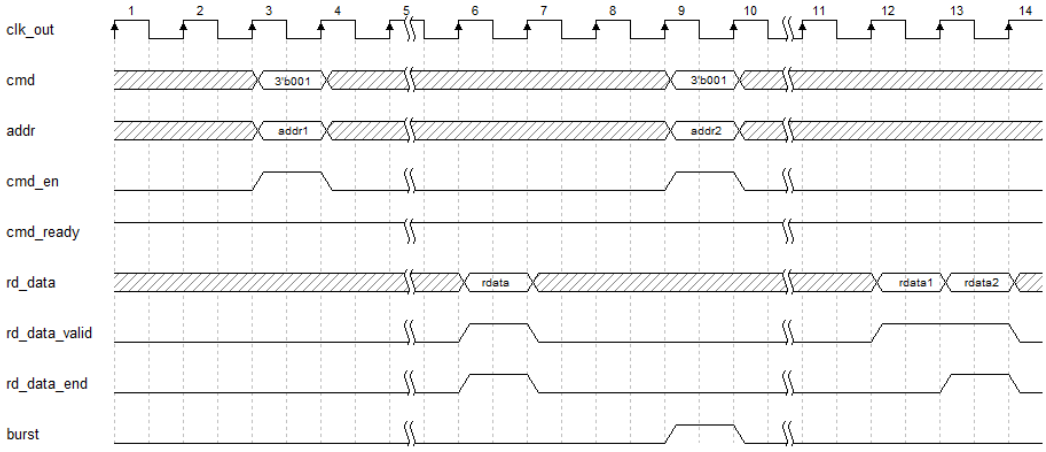
Similar to the write operation, when clock ratio is 1:2 and users configure the burst length with BL8, read data occupies two clk cycles, as shown in Figure 4-15.

Figure 4-15 Read Data Timing with 1:2 Clock Ratio and Burst Mode=BL8



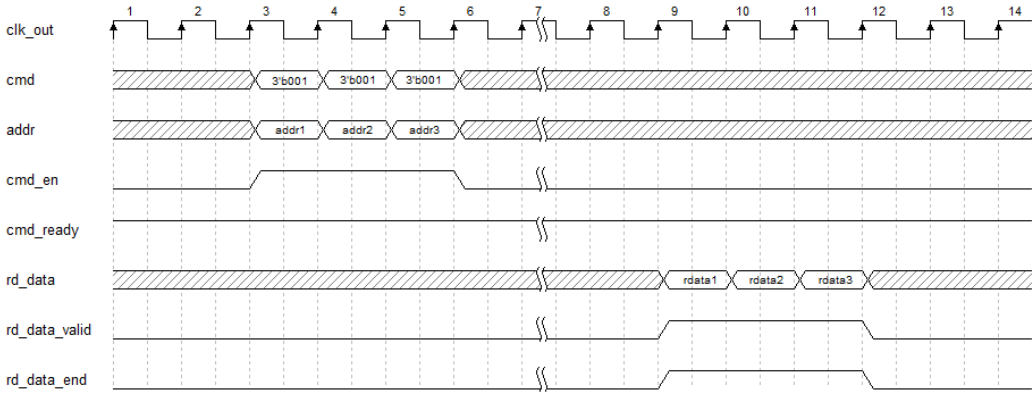
When clock ratio is 1:2 and users configure the burst mode with OTF (i.e., users can control the burst port to switch BC4 and BL8), when cmd_en is valid, burst 0 indicates read BC4 and the read data occupies one clk cycle; burst 1 indicates read BL8 and the read data occupies two clk cycles, as shown in Figure 4-16.

Figure 4-16 Read Data Timing with 1:2 Clock Ratio and Burst Mode= OTF



When the clock ratio is 1:4, the read data is returned sequentially according to the read command order, as shown in Figure 4-17.

Figure 4-17 Read Data Timing in Non-continuous Burst Mode with 1:4 Clock Ratio



4.4.9 Refresh

Refresh

The DDR3 SDRAM array needs to continually refresh to ensure the data is not lost. Therefore, Gowin DDR3 Memory Interface IP is required to send the refresh instruction to DDR3 SDRAM periodically. Gowin DDR3 Memory Interface IP requires ad interval of t_{REFI} time to generate refresh command; after the refresh command is generated, MC will precharge all banks after executing the last read/write command, and then execute Refresh command. The refresh operation has higher priority.

Self-Refresh

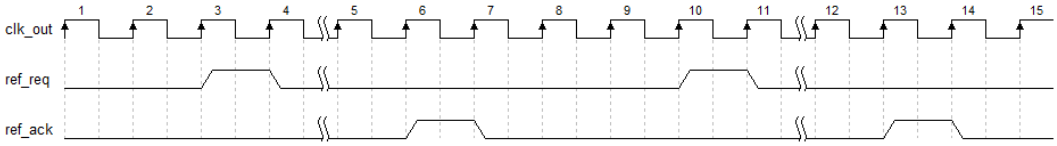
You can send a self-refresh request through the sr_req port and configure DDR3 SDRAM with self-refresh mode. In the self-refresh mode, DDR3 SDRAM will automatically refresh the memory array without a refresh instruction from Gowin DDR3 Memory Interface IP.

Gowin DDR3 Memory Interface IP will return the sr_ack signal to users after the self-refresh configuration of DDR3 SDRAM is completed, indicating that the DDR3 SDRAM is configured with self-refresh mode.

User Refresh

You can use this option to configure the refresh mode with user refresh mode. "USER_REFRESH" needs to be configured with "ON." You can send a refresh request to MC through ref_req port at any time without additional commands to MC through cmd port. When user refresh instruction is sent, ref_req need to hold at high level for one cycle. After the refresh instruction is sent, MC will feedback ref_ack signal which will hold at high level for one cycle, as shown in Figure 4-18.

Figure 4-18 User Refresh Timing



The user refresh operation may affect the instructions that MC has received or is executing. Before user refresh operation, MC completes the command operation being executed first. You should consider the coordination between the user refresh and other instructions to avoid t_{REFI} violation.

Considering the worst case, you can operate user refresh with reference to the following formula. Command a certain time to complete the transmission, which can be roughly calculated by parameters such as t_{RCD}, CL, data transmission time, t_{RP}, etc. And the user refresh should be completed before the time parameter t_{REFI} is violated, so the maximum interval between two user refresh times can be calculated by following formula.

$$t_{REFI} - (t_{RCD} + (CL + 4) \times t_{CK} + t_{RP}) \times nBA$$

In application, if a user refresh starts, you need to start a user refresh immediately after the DDR3 SDRAM completes initialization to establish the time base for subsequent user refresh requests.

4.5 AXI4 Interface

The memory controller supports the industry-standard AXI4 interface. The AXI4 slave interface module maps AXI4 transaction information to the internal memory control logic. The AXI4 slave interface is optional. Users can enable or disable generation of the AXI4 slave interface through the "AXI4 Interface" option in the IP configuration GUI. For detailed protocol information on the AXI4 interface, refer to the Arm AMBA AXI Specifications.

All AXI4 interface parameters can be configured through the "AXI Interface Options" in the IP GUI.

4.5.1 Arbitration Mode

The AXI4 write and read channels support five arbitration modes, which are described as follows:

- Mode 0: Read and write channels have equal priority. The arbiter

alternates between the read and write channels on every clock cycle.

- Mode 1: Read and write channels have equal priority. When both channels are idle, the arbiter continuously selects the write channel. If read and write requests occur simultaneously, the write request is serviced first. If the last transaction issued by the master was a write, the arbiter gives priority to the read channel in the next arbitration; if the last transaction was a read, the arbiter gives priority to the write channel.
- Mode 2: Read-priority mode. When read and write requests occur simultaneously, the read request is serviced first.
- Mode 3: Write-priority mode. When read and write requests occur simultaneously, the write request is serviced first.
- Mode 4: Read and write channels have equal priority. The arbiter switches to the write channel when either of the following conditions is met:
  - Only write requests are present and no read requests exist.
  - Both read and write requests are present, but the read channel is blocked and cannot continue. If the blocking duration reaches 256 system clock cycles, the arbiter switches to the write channel.

Note!

The conditions for switching to the read channel are symmetrical to those for switching to the write channel.

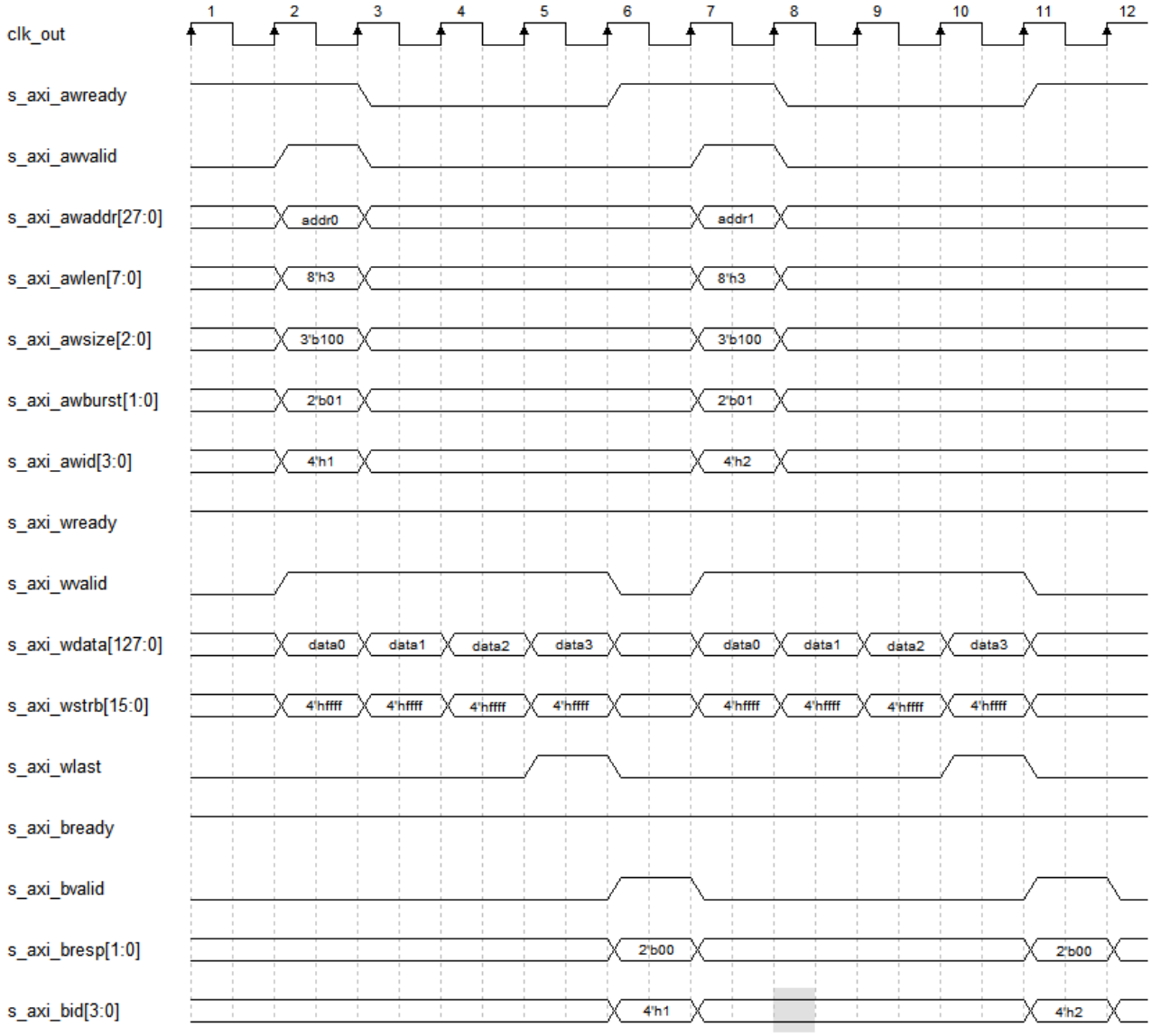
- Mode 5: Read-priority mode. When both read and write requests are present, the arbiter prioritizes the read channel. The write channel starts a wait timer, and if the wait time reaches 256 system clock cycles, the arbiter forcibly switches to the write channel.

4.5.2 AXI4 Interface Timing

AXI4 Write Channel Timing

When the write burst length is 4 and the burst type is 2'b01 (INCR), the timing is shown in Figure 4-19.

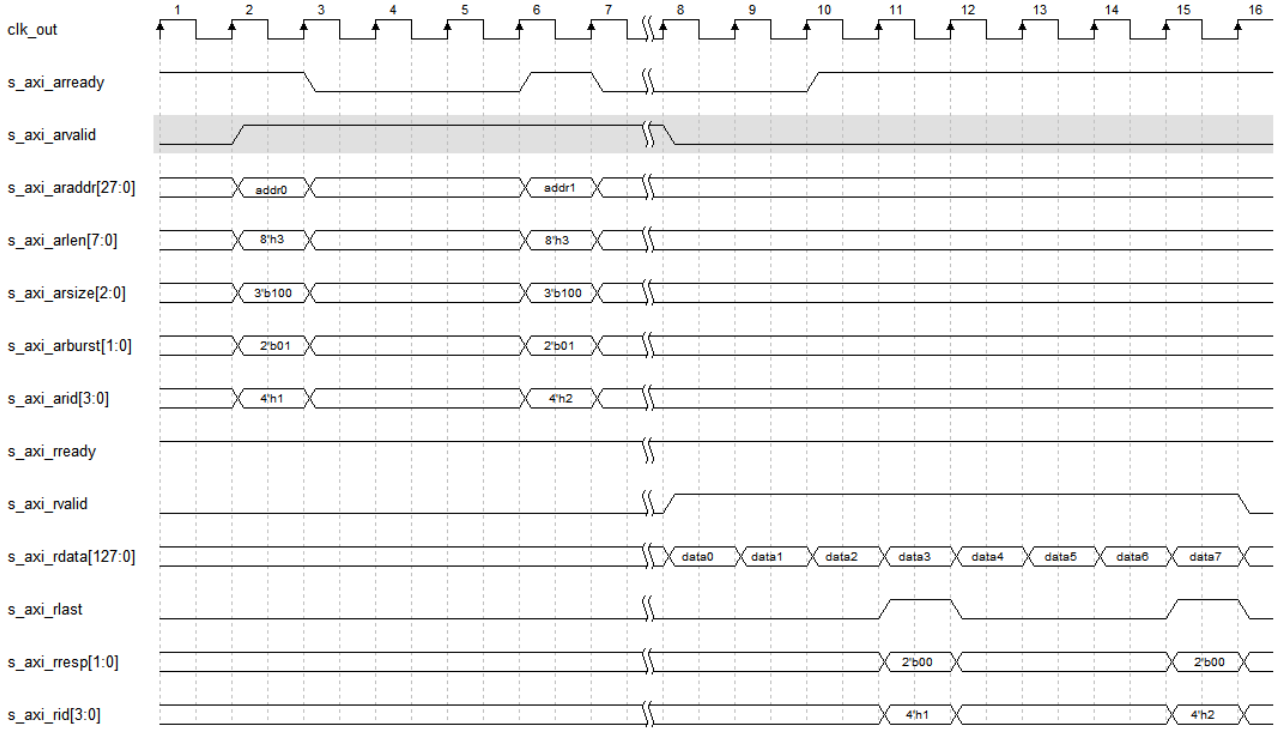
Figure 4-19 AXI4 Write Channel Timing Diagram



AXI4 Read Channel Timing

When the read burst length is 4 and the burst type is 2'b01 (INCR), the timing is shown in Figure 4-20.

Figure 4-20 AXI4 Read Channel Timing Diagram



5 ODT

On-Die Termination (ODT) is an important function of DDR3 that can help reduce signal reflections on DQS, DQ, and DM lines, thereby improving signal quality. Both the FPGA and DDR3 chips support ODT function. When the FPGA sends DQS, DQ, and DM signals to the DDR3 chip, the IP activates ODT on the DDR3 side by driving the ODT pin high, while the ODT on the FPGA side is automatically disabled. Conversely, when the DDR3 chip sends DQS and DQ signals to the FPGA, the IP the IP deactivates ODT on the DDR3 side by driving the ODT pin low, and the ODT on the FPGA side is automatically enabled.

It is not always necessary to enable ODT on the FPGA side, as enabling it increases power consumption. In practical applications, users can try operating without enabling ODT on the FPGA side. If DDR3 initialization fails or data errors occur, ODT on the FPGA side can then be enabled for debugging.

In Gowin Software, ODT is disabled by default. Users can enable it via the FloorPlanner tool. Taking GW5A series of FPGA as an example, the method for enabling ODT is described below. As shown in Figure 5-1, set the Single Resistor of DQS and DQ to 50. If the setting of 50 does not yield good results, users can try other values. Gowin Software enables the Diff Register of DQS by default, and users need to disable it.

Figure 5-1 ODT Configuration in FloorPlanner

Constraints

Port	Direction	Diff Pair	Location	Bank	Exclusive	IO Type	Drive	Pull Mode	PCI Clamp	Hysteresis	Open Drain	Vref	Single Resistor	Diff Resistor
ddr_dqs[3]	inout	ddr_dqs_n[3]	N15,N16	3	False	SSTL15D_I	8	NONE	OFF	NONE	OFF	OFF	50	OFF
ddr_dqs[2]	inout	ddr_dqs_n[2]	K17,K18	3	False	SSTL15D_I	8	NONE	OFF	NONE	OFF	OFF	50	OFF
ddr_dqs[1]	inout	ddr_dqs_n[1]	J13,K14	2	False	SSTL15D_I	8	NONE	OFF	NONE	OFF	OFF	50	OFF
ddr_dqs[0]	inout	ddr_dqs_n[0]	D17,D18	2	False	SSTL15D_I	8	NONE	OFF	NONE	OFF	OFF	50	OFF
ddr_dq[9]	inout		K12	2	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[8]	inout		G16	2	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[7]	inout		C18	2	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[6]	inout		F16	2	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[5]	inout		C17	2	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[4]	inout		H13	2	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[3]	inout		F14	2	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[31]	inout		U18	3	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[30]	inout		N18	3	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[2]	inout		E16	2	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[29]	inout		U17	3	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF
ddr_dq[28]	inout		R18	3	False	SSTL15_I	8	NONE	OFF	NONE	OFF	INTERNAL	50	OFF

The ODT of the DDR3 chip is configured via the Rtt Nom and Rtt Wr options. In multi-rank configurations, both Rtt Nom and Rtt Wr need to be set in order to use DDR3 Dynamic ODT. In single-rank configuration, only Rtt Nom needs to be set. It is important to note that if only Rtt Wr is configured and Rtt Nom is not, when the ODT pin is driven high, the DDR3 chip will internally use Rtt Wr as its ODT value.

6 Port List

The I/O ports of Gowin DDR3 Memory Interface IP are as shown in Table 6-1. For I/O ports of AXI4 interface, see Table 6-2.

Table 6-1 IO Port List of Gowin DDR3 Memory Interface IP

Signal	Data Width	I/O	Description
User Interface			
addr	ADDR_WIDTH	Input	Address input, configure parameter to set the signal width
cmd	3	Input	Command channel
cmd_en	1	Input	Command and address enable signals 0: invalid 1: valid
cmd_ready	1	Output	High indicates Memory Interface can receive commands and addresses.
rd_data	APP_DATA_WIDTH	Output	Read data channel
rd_data_end	1	Output	High indicates the end cycle of the current rd_data.
rd_data_valid	1	Output	rd_data valid signal: 0: invalid 1: valid
burst	1	Input	OTF control port, BL8 mode when 1'b1; BC4 mode when 1'b0, valid only in OTF mode
wr_data	APP_DATA_WIDTH	Input	Write data channel
wr_data_end	1	Input	High level indicates that the current clock cycle is the last cycle of this wr_data.
wr_data_mask	APP_MASK_WIDTH	Input	Each bit corresponds to one byte of wr_data. 0: The corresponding wr_data byte mask is invalid.

Signal	Data Width	I/O	Description
			1: The corresponding wr_data byte mask is valid.
wr_data_rdy	1	Output	High indicates MC can receive the user data.
wr_data_en	1	Input	wr_data write enable signal: 0: Disable 1: Enable
sr_req	1	Input	Self-refresh request
sr_ack	1	Output	Refresh the acknowledgement signal
ref_req	1	Input	User refresh request
ref_ack	1	Output	User refresh the acknowledgement signal
clk	1	Input	Reference input clock, generally PCB crystal input, 50M recommended.
memory_clk	1	Input	Memory interface frequency input by user; when GW2A devices used, this clock can be either the output clock of PLL or other clock; when GW5A devices used, this clock must be output from clkout2 of PLL.
pll_stop	1	output	For the details, you can see 4.4.4 pll_stop.
pll_lock	1	Input	If memory_clk is PLL multiplication input, this interface is connected to the pll_lock of PLL, if you do not use PLL, this interface is connected to high level.
rst_n	1	Input	System reset input signal 0: Enable 1: Disable
init_calib_complete	1	Output	Initialization completed signal
clk_out	1	Output	User design clock
ecc_err	APP_DATA_WIDTH/32	Output	ECC indicates signal output
ddr_rst	1	Output	The reset signal that IP processed is used for user design, high reset.
DDR3 SDRAM Interface			
O_ddr_addr	ROW_WIDTH	Output	Row address (activation command), Column address (read/write command)
O_ddr_bank	BANK_WIDTH	Output	Bank address

Signal	Data Width	I/O	Description
O_ddr_cs_n	CS_WIDTH	Output	Chip selected, active-low.
O_ddr_ras_n	1	Output	Row address selection signal
O_ddr_cas_n	1	Output	Column address selection signal
O_ddr_we_n	1	Output	Row write enable
O_ddr_ck	CK_WIDTH	Output	A clock signal provided to DDR3 SDRAM
O_ddr_ck_n	CK_WIDTH	Output	Compose a differential signal with ddr_ck
O_ddr_cke	CKE_WIDTH	Output	DDR3 SDRAM clock enable signal
O_ddr_odt	ODT_WIDTH	Output	Terminating resistor control of memory signal
O_ddr_reset_n	1	Output	DDR3 SDRAM reset signal
O_ddr_dm	DM_WIDTH	Output	DDR3 SDRAM data masking signal
IO_ddr_dq	DQ_WIDTH	Bidirection	DDR3 SDRAM data
IO_ddr_dq	DQS_WIDTH	Bidirection	DDR3 SDRAM data strobe signal
IO_ddr_dqs_n	DQS_WIDTH	Bidirection	Compose a differential signal with ddr_dqs

The I/O ports of AXI4 interface are described as follows:

Table 6-2 Port List of AXI4 Interface

Signal Name	Width	Direction	Description
s_axi_awvalid	1	input	Write address valid signal
s_axi_awready	1	output	Write address ready signal. When high, the slave is ready to receive write address control signals.
s_axi_awid	ID Width	input	Write address ID
s_axi_awaddr	Address Width	input	Write address
s_axi_awlen	8	input	Write burst length
s_axi_awsize	3	input	Size of each transfer in the burst. Must be 1/8 of s_axi_wdata width.
s_axi_awburst	2	input	Write burst type. Only INCR and WRAP are supported.
s_axi_wvalid	1	input	Write data valid signal
s_axi_wready	1	output	Write data ready signal. When high, the slave is ready to receive write data.
s_axi_wdata	Data Width	input	Write data
s_axi_wstrb	Data Width/8	input	Write data strobe (mask)
s_axi_wlast	1	input	Write data last signal. High indicates the last data of a write

Signal Name	Width	Direction	Description
			transfer.
s_axi_bvalid	1	output	Write response valid signal
s_axi_bready	1	input	Write response ready signal
s_axi_bresp	2	output	Write response
s_axi_bid	ID Width	output	Write response ID
s_axi_arvalid	1	input	Read address valid signal
s_axi_arready	1	output	Read address ready signal. When high, the slave is ready to receive read address control signals.
s_axi_arid	ID Width	input	Read address ID
s_axi_araddr	Address Width	input	Read address
s_axi_arlen	8	input	Read burst length
s_axi_arsize	3	input	Size of each transfer in the burst. Must be 1/8 of s_axi_wdata width.
s_axi_arburst	2	input	Read burst type. Only INCR and WRAP are supported.
s_axi_rvalid	1	output	Read data valid signal
s_axi_rready	1	input	Read data ready signal
s_axi_rdata	Data Width	output	Read data
s_axi_rresp	2	output	Read response
s_axi_rid	ID Width	output	Read response ID
s_axi_rlast	1	output	Read data last signal. High indicates the last data of a read transfer.

7 Parameter Configuration

7.1 Gowin DDR3 Memory Interface IP Parameters

Gowin DDR3 Memory Interface IP supports DDR3 SDRAM devices. Users shall configure the static parameters and timing parameters of the Gowin DDR3 Memory Interface IP according to the design requirements. The static parameters are described as follows.

- User Interface: The user interface allows you to select either Controller or PHY. When Controller is selected, the IP includes both Controller and PHY functions. When "PHY" is selected, the IP only includes the PHY function, and the user is required to implement the controller function independently.
- Memory Type: Supports configuration of four memory types: Components, RDIMMs, UDIMMs, and SODIMMs.
- Memory Clock: Used to set the high-speed clock frequency for the IP. The configured frequency should match the clock frequency corresponding to the DDR3 interface rate. For example, if the DDR3 rate reaches to 800 Mbps, this option should be set to 400 MHz.
- CLK Ratio: Used to define the ratio between the Controller and the PHY. Options are 1:4 or 1:2. The rate of the PHY is the same as the rate of the Memory Clock, while the rate of the Controller is 1/4 or 1/2 of the Memory Clock rate.
- Dq Width: The total DQ width. For instance, if a single DDR3 chip has a DQ width of 16 bits and the IP drives two such chips, then the Dq Width should be 32 bits.
- Dram Width: The DQ width of a single DDR3 chip.
- Rank Address: Fixed to 1. The IP does not support the rank address, so this value does not take effect.
- Bank Address: Fixed to 3.
- Row Address: Row address width. The user should refer to the datasheet of the selected DDR3 chip to configure this value.
- Column Address: Column address width. The user should refer to the datasheet of the selected DDR3 chip to configure this value.

- SIM: Should be enabled during simulation to shorten DDR3 initialization time; disabled in non-simulation environments.
- Address Width: Read-Only; user interface address bus width, calculated as Rank Address + Bank Address + Row Address + Column Address. The most significant bit (Rank Address) is unused and must be fixed to 0.
- Data Width: Read-Only; user interface data bus width; When CLK Ratio is set to 1:4, the width equals Dq Width* 8; when it is set to 1:2, the width equals Dq Width*4.
- Mask Width: Read-Only; user interface data mask bus width; when CLK Ratio is set to 1:4, the width equals Dq Width*8/8; when it is set to 1:2, the width equals Dq Width*4/8.
- Burst Mode: Used to configure the burst mode of the DDR3 chip. Supports BC4 and BL8.
- Burst Type: Used to configure the burst type of the DDR3 chip. Supports Sequential and Interleaved. For detailed descriptions of Sequential and Interleaved, please refer to the JESD79-3F specification.
- CAS Latency (CL): The delay from a read command to the first valid data. The total read latency (RL) is calculated as: Additive Latency (AL) + CAS Latency (CL).
- CW Latency (CWL): The delay from a write command to the first valid data. The total write latency (WL) is calculated as: Additive Latency (AL) + CW Latency (CWL).
- Additive Latency (AL): Supports configuration of 0, CL−1, and CL−2. DDR3 supports the Posted CAS, which can improve efficiency when there is a timing conflict between Active and Read/Write commands.
- Write Recovery: Refers to the number of clock cycles that must be waited after completing a valid write operation in an activated bank before issuing a precharge command.
- Rtt Nom, Rtt Wr: DDR3 On-Die Termination (ODT) settings. Rtt_Nom supports OFF, 20Ω, 30Ω, 40Ω, 60Ω, and 120Ω. Rtt_Wr supports OFF, 60Ω, and 120Ω.
- OUTPUT DRV: Used to set the output drive strength for DQS and DQ signals. "LOW" corresponds to an impedance of RZQ (240Ω)/6=40Ω. "HIGH" corresponds to an impedance of RZQ (240Ω)/7=34Ω. Changing the OUTPUT DRV setting affects signal quality. If DDR3 initialization fails or data read/write errors occur, you can try modifying this option for debugging.
- Debug Parameter, Debug Port: These are intended for engineering debug use. Users do not need to configure or be concerned with these settings.

You also need to configure the timing parameters of Gowin DDR3 Memory Interface IP according to the design requirements, and the timing

parameter descriptions are as shown in Table 7-1.

Table 7-1 DDR3 Time Parameter

Name	Description
tCK	Memory interface clock period (ps)
tCKE	Minimum pulse time of CKE signal (ps)
tFAW	The interval between simultaneous activation commands of more than four lines in the same rank is allowed, so the minimum value should be no less than four times the tRRD.
tRAS	Time from ACTIVE to PRECHARGE
tRCD	Time from ACTIVE to READ/WRITE
tREFI	Memory refresh interval
tRFC	The interval from REFRESH to ACTIVE/REFRESH
tRP	PRECHARGE cycle
tRRD	The interval from ACTIVE to ACTIVE between different banks
tRTP	The interval from READ to PRECHARGE
tWTR	The interval from WRITE to READ
AL	Additive latency
CL	CAS latency

7.2 AXI4 Interface Parameters

Data Width

Data Width refers to the bit widths of the AXI4 write data bus and read data bus, read-only. When the CLK Ratio is set to 1:4, the data width equals Dq Width × 8; when the CLK Ratio is set to 1:2, the data width equals Dq Width × 4. When ECC is enabled, Dq Width is fixed at 64. When ECC is disabled, Dq Width equals the actual total DQ width.

Address Width

Address width refers to the bit width of the read and write address buses, read-only. The AXI4 interface uses byte addressing, whereas the DDR3 address is not necessarily byte-addressed. For example, when Dq Width is set to 8, the DDR3 address corresponds to a byte address, and in this case, the AXI4 interface address and the DDR3 address are equivalent. When Dq Width is set to 16, one DDR3 address stores two bytes of data; the AXI4 interface address and the DDR3 address widths are not equal. Therefore, the AXI4 address bus width differs from the DDR3 address width.

ID Width

ID Width refers to the bit width of all IDs (AWID, BID, ARID, and RID). This parameter is configurable, with a valid range from 1 to 16.

AXI Buffer Type

Specifies the FPGA resources used for buffers within the AXI4 interface. Supported options include BSRAM, SSRAM, and REG.

Arbitration Mode

For the descriptions, you can see section 4.5.1 Arbitration Mode.

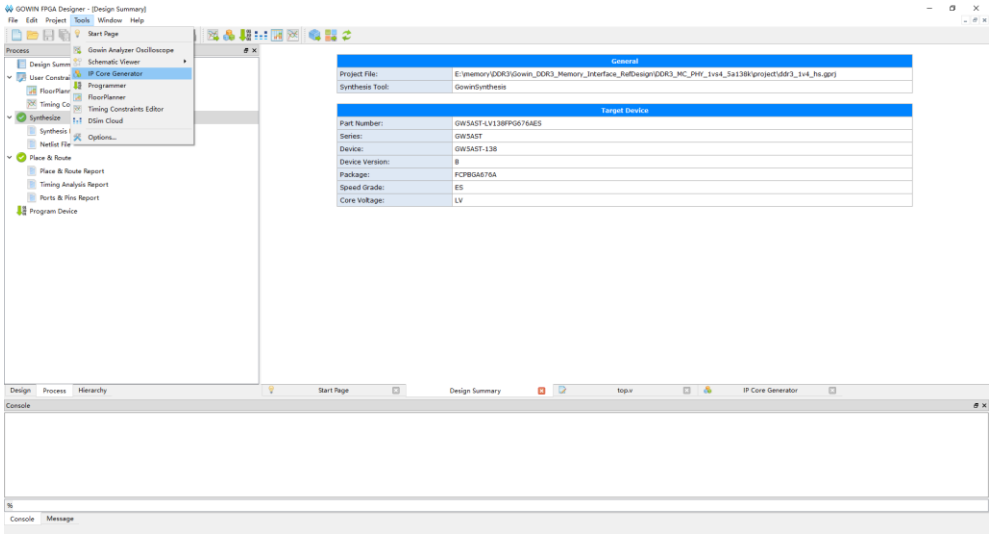
8 Interface Configuration

You can invoke and configure the Gowin DDR3 Memory Interface IP using the IP Core Generator tool in the IDE. This chapter takes AD3U160022G11 memory chip as an example, introduces the main configuration interface, configuration flow and the meaning of each configuration option (Take the clock ratio of 1:2 as an example).

1. Open IP Core Generator.

After creating the project, click the "Tools" tab in the upper left, click “IP Core Generator” to open Gowin IP Core Generator via the drop-down list, as shown in Figure 8-1.

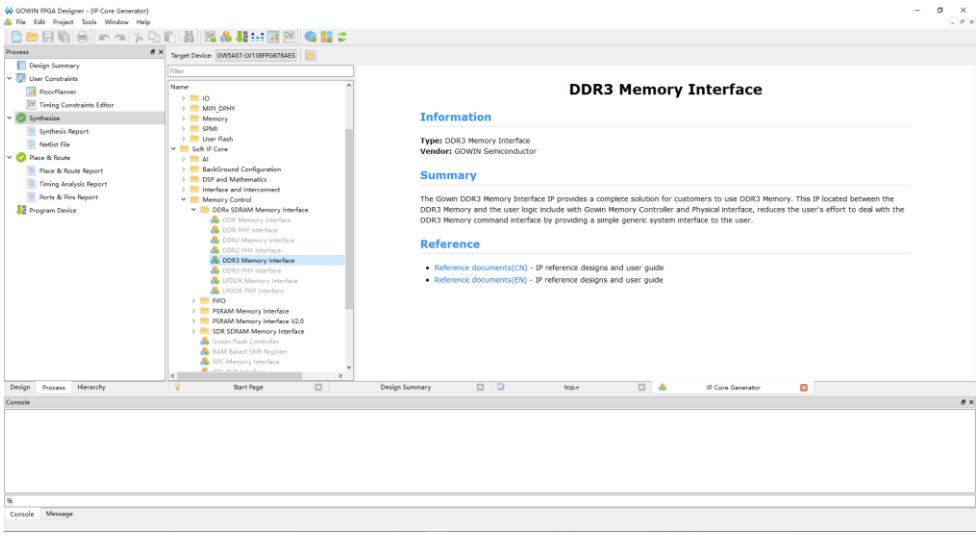
Figure 8-1 Open IP Core Generator



2. Open the DDR3 Memory Interface IP core.

Click DDR option, double-click DDR3 Memory Interface. The configuration interface of DDR3 Memory Interface IP core is as shown in Figure 8-2.

Figure 8-2 Open DDR3 Memory Interface IP Core

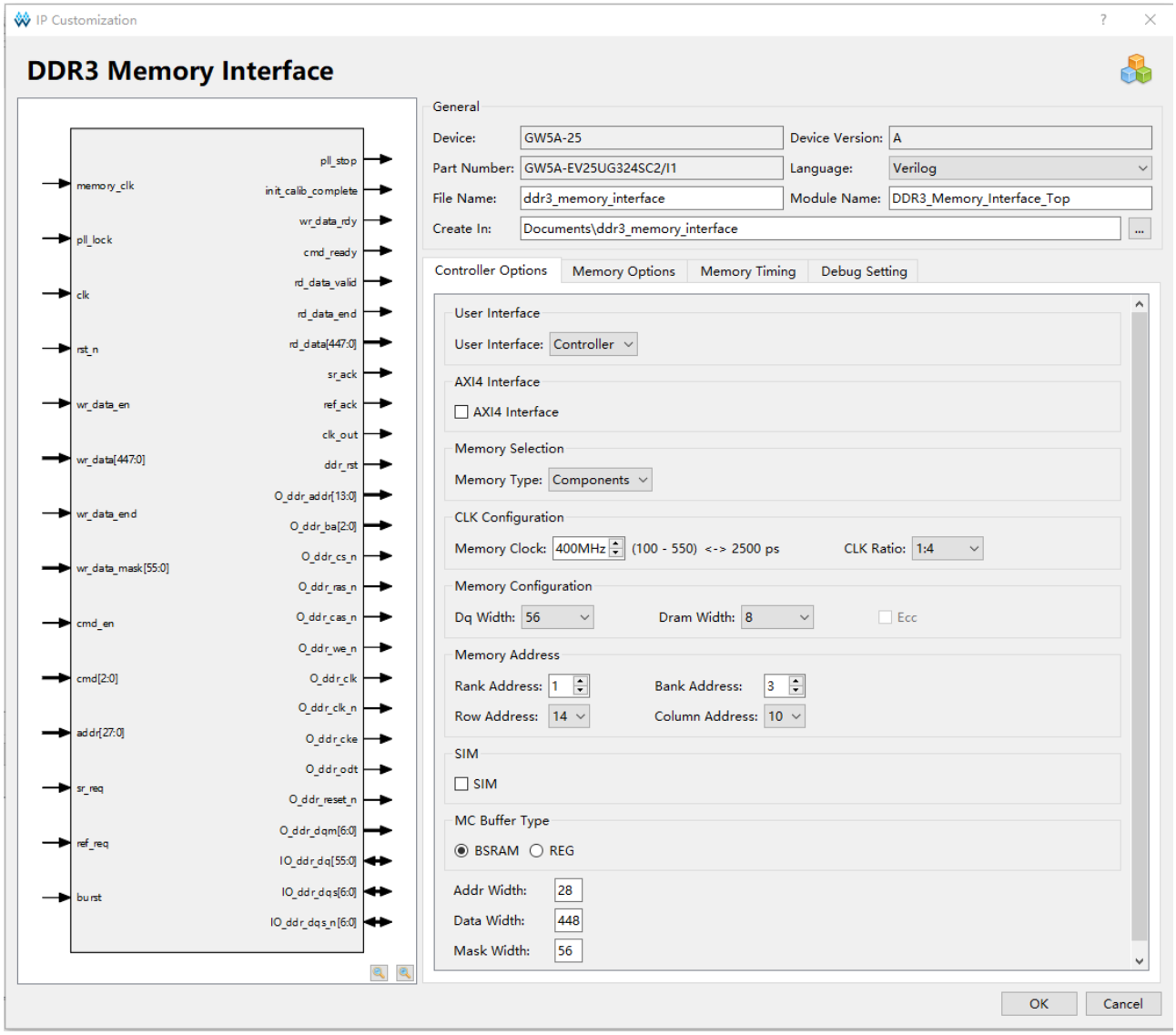


3. DDR3 Memory Interface IP Core Interface.

The port diagram of the DDR3 Memory Interface IP core is on the left, as shown in Figure 8-3.

The left is the ports of DDR3 Memory Controller and the user end. You can connect your design to the DDR3 Memory Interface IP to send/receive commands and data. The right is the interface of the PHY and DDR3 chip. You can save and access data by connecting the DDR3 Memory Interface IP and DDR3 chip. With different configuration information, the signal bit width and the signal number will change accordingly.

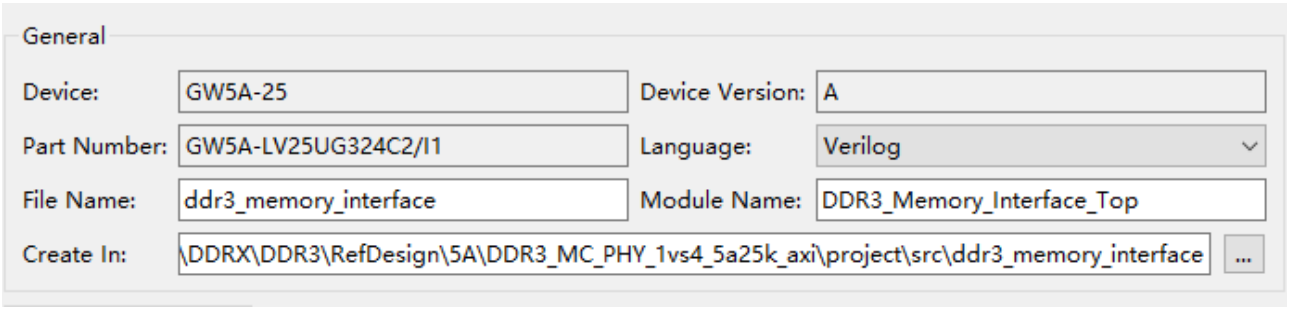
Figure 8-3 IP Core Interface



4. Configure General Information.

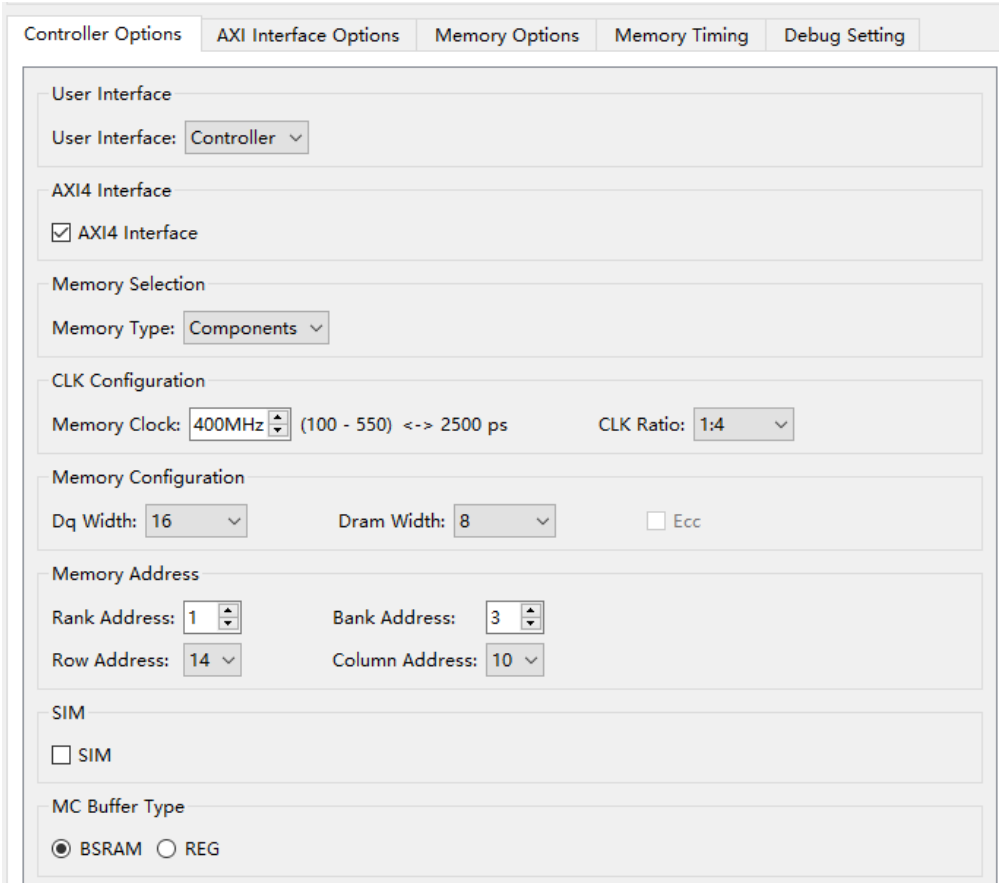
See the general information in the upper part of the configuration interface. The "Module Name" displays the top-level file name of the generated project, and the default is "ddr3_memory_interface_top", and you can modify the name. The "File Name" displays the folder generated by the IP core, which contains the files required by DDR3 Memory Interface IP core, and the default is "DDR3 Memory_Interface", and you can modify the path. "Create In" displays the path of IP core folder. The default is "\project path\src\DDR3_Memory_Interface", and you can modify the path, as shown in Figure 8-4.

Figure 8-4 Basic Information Configuration Interface



5. The parameter configuration interface is as shown in Figure 8-5. For details of the parameter configuration, you can see chapter 7 Parameter Configuration.

Figure 8-5 Parameter Configuration




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